

through decentralised models, where the contribution of all developers impacts the overall direction of a project. Therefore, without a clearly defined governance framework, there is a high probability of conflict, delays in the decision-making process, and inconsistent involvement of the project. In addition to the complicated nature of collaborative project management, the scalability of repository management can also prove troublesome when trying to develop a large blockchain project with a lot of different components, including smart contracts, front-end applications, APIs, and infrastructure scripts. Therefore, as a project increases in size, it becomes more difficult to manage complex repositories while also having the appropriate version controls and collaborative efforts across contributors.

Open source blockchain development also has a risk of code misuse and the creation of unauthorised forks, as public access to code allows any individual to copy, alter, or deploy a similar asset. These actions affect the credibility of the original asset while also adding to the confusion of the entire ecosystem. Using external tools and integrations also adds another level of complexity to open source blockchain development. Workflow processes related to blockchain require multiple frameworks and methodologies to test and deploy assets. Therefore, as the integration of these external tools and processes into GitHub-based workflows takes place, continuous monitoring and regular updates across the integration are required to ensure compatibility and stability.

The future

Blockchain coding on GitHub is likely to make major strides due to the development of decentralised technology and broader developer communities. Emerging technologies will lead to GitHub's strategic use as

both a communication and a cognitive layer to facilitate the creation and verification of digitally distributed systems. AI will have a significant influence on the integration of digital technologies into the development of blockchain applications. The use of AI-enabled coding tools will provide smart contract template generation capabilities, detect/identify logical flaws in smart contracts, and provide techniques to improve code structure efficiency. These tools will assist developers and improve productivity by decreasing the hazards associated with manual coding errors in the blockchain space.

A key opportunity for automated security and verification mechanisms to evolve is their integration with the GitHub development pipeline. The use of advanced testing frameworks, formal verification tools, and constant audit pipelines will improve the security and reliability of smart contracts prior to their deployment, solving one of the major challenges faced in the development of blockchain technology. The impact of decentralised development ecosystems may also alter GitHub's future role as well as peer-to-peer networks on distributed version control (e.g., Git) systems, which were created to reduce dependency

on centrally hosted services. If GitHub were to be incorporated into a network of decentralised storage and collaborating technologies, it could change the way developers create and maintain their blockchain projects over time.

The aim is for GitHub to serve as the main identity platform for developers building on the blockchain. The more developers evaluate their own experiences and contributions to the community, including things like their contribution history, activity in the various repositories, and involvement in community-based projects, the more they will have an established level of credibility and be able to leverage the many global opportunities in Web3.

With the increase in adoption of blockchain technology, there will be increased needs for standardisation of the methodologies used to develop software. Creating similar methods around how to structure repositories, and how to test and deploy all pieces of development by teams, will improve collaboration among GitHub projects. This evolution will support the overall goal of making technology more accessible, inclusive, and decentralised. **END** 🐧

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By: Dhruv Panchal

The author is an AI engineer and researcher.